

Redox

Redox reactions · Galvanic cells · Corrosion · Electrolysis

Completed revision summary: definitions of redox, oxidation numbers, balancing half-equations in acidic solution, combining half-equations, oxidants & reductants, standard reduction potentials, galvanic (electrochemical) cells, commercial cells, corrosion & its prevention, and electrolysis with its industrial applications. All half-equation balancing shown in **acidic** conditions only.

Redox fundamentals

- 1 Definitions of redox
- 2 Oxidation numbers
- 3 Using oxidation numbers (half-equations)
- 4 Net redox from two half-equations
- 5 Terminology: oxidants & reductants
- 6 Standard reduction potentials

Cells

- 7 Galvanic (electrochemical) cells

- 8 Commercial galvanic cells

- 9 Corrosion & its prevention

Electrolysis

- 10 Electrolysis

- 11 Applications of electrolysis

Reference

- R Standard reduction potential data

REDOX FUNDAMENTALS

1 Definitions of Redox

A **redox reaction** is one in which electrons are transferred. Oxidation and reduction always occur **together** — the electrons lost by one species are gained by another.

Redox as the transfer of electrons

ELECTRON DEFINITION (THE MODERN, GENERAL ONE)

- **Oxidation** = **loss** of electrons.
- **Reduction** = **gain** of electrons.

OIL RIG

Oxidation Is Loss · Reduction Is Gain (of electrons)

Redox as the transfer of oxygen

OXYGEN DEFINITION (HISTORICAL)

- **Oxidation** = **gain** of oxygen. e.g. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (Mg is oxidised).
- **Reduction** = **loss** of oxygen. e.g. $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$ (CuO is reduced).

Redox as the transfer of hydrogen

HYDROGEN DEFINITION (HISTORICAL)

- **Oxidation** = **loss** of hydrogen. e.g. $\text{H}_2\text{S} \rightarrow \text{S} + 2\text{H}^+ + 2\text{e}^-$ (S is oxidised).
- **Reduction** = **gain** of hydrogen. e.g. $\text{Cl}_2 + \text{H}_2 \rightarrow 2\text{HCl}$ (Cl is reduced).

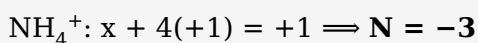
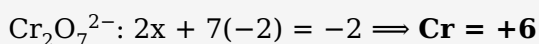
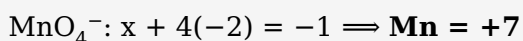
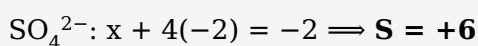
2 Oxidation Numbers

The **oxidation number** is the charge an atom *would* have if every bond were completely ionic. It is the tool used to track electron transfer.

RULES FOR ASSIGNING OXIDATION NUMBERS

- Elements are always **0** (e.g. Na, O₂, Cl₂, S₈).
- Monatomic ions are always **equal to the charge of the ion** (Na⁺ = +1, S²⁻ = -2, Al³⁺ = +3).
- Hydrogen is always **+1** ...
- ... except in **metal hydrides** where it is **-1** (e.g. NaH, CaH₂).
- Oxygen is always **-2** ...
- ... except in **peroxides** where it is **-1** (e.g. H₂O₂), and in **OF₂** where it is +2.
- In a **polyatomic ion**, the sum of all oxidation numbers must equal **the charge on the ion**.
- In a **compound**, the sum of all oxidation numbers must equal **zero**.

WORKED EXAMPLES



3 Using Oxidation Numbers

Identifying a redox reaction

HOW OXIDATION NUMBERS SHOW A REACTION IS REDOX

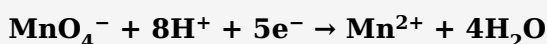
If the oxidation number of **any** element **changes** between reactants and products, the reaction is redox. An **increase** = oxidation; a **decrease** = reduction. If no oxidation number changes, it is **not** a redox reaction.

Balancing a redox half-equation — acidic conditions only

THE FOUR STEPS (ACIDIC SOLUTION)

- **Step 1:** Balance all atoms *except* O and H (the atom being oxidised/reduced first).
- **Step 2:** Balance **oxygen** by adding **H₂O**.
- **Step 3:** Balance **hydrogen** by adding **H⁺**.
- **Step 4:** Balance **charge** by adding **electrons (e⁻)**; then check atoms and charge both balance.

EXAMPLE — PERMANGANATE REDUCED IN ACID



Check: atoms balance (1 Mn, 4 O, 8 H each side); charge = (-1 + 8 - 5) = +2 on the left = +2 on the right ✓

4 Writing Net Redox Reactions from Two Half-Equations

What half-equations show

A half-equation shows **one** process on its own — either an oxidation (electrons on the right, lost) or a reduction (electrons on the left, gained) — together with the electrons involved.

Using the Standard Reduction Potentials table

READING THE DATA BOOKLET TABLE TO GET COMMON HALF-EQUATIONS

- Read **left to right** (as printed) → a **reduction** half-equation (the species gains electrons).
- Read **right to left** (reversed) → an **oxidation** half-equation (the species loses electrons); reverse the sign meaning of E° when it acts this way.

Combining two half-equations into a net redox equation

THE FOUR STEPS

- **Step 1:** A **reduction** and an **oxidation** half-equation must both be present (reverse one so one is written each way).
- **Step 2:** Multiply each half-equation so the **number of electrons is equal** in both.
- **Step 3:** **Add** the two half-equations together.
- **Step 4:** **Cancel** the electrons and any species identical on both sides (e.g. H^+ , H_2O); check atoms and charge.

EXAMPLE — MnO_4^- OXIDISING Fe^{2+} IN ACID

Reduction ($\times 1$): $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Oxidation ($\times 5$): $5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$

Net: $MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

Charge check: left +17 = right +17 ✓

5 Terminology: Oxidants & Reductants

Oxidant / oxidising agent

DEFINITION

An **oxidising agent (oxidant)** causes another species to be oxidised by **accepting/gaining electrons**; the oxidant is itself **reduced**.

Common oxidising agents: acidified permanganate (MnO_4^- , purple $\rightarrow$ colourless Mn^{2+}), acidified dichromate ($Cr_2O_7^{2-}$, orange $\rightarrow$ green Cr^{3+}), the halogens (F_2 , Cl_2 , Br_2), hydrogen peroxide (H_2O_2), oxygen (O_2), concentrated nitric acid (HNO_3) and Fe^{3+} .

WHICH ACID IS USED TO ACIDIFY THE OXIDISING AGENT — AND WHY?

Dilute sulfuric acid (H_2SO_4). It supplies the H^+ ions needed, but its ions are **not easily oxidised or reduced**, so it does not interfere with the reaction.

- **Not HCl** — the chloride ion Cl^- can itself be oxidised (e.g. by MnO_4^- or $\text{Cr}_2\text{O}_7^{2-}$) to Cl_2 , interfering with the result.
- **Not (dilute) HNO_3** — the nitrate ion is itself an oxidising agent and would compete with the oxidant being studied.

Reductant / reducing agent

DEFINITION

A **reducing agent (reductant)** causes another species to be reduced by **donating/losing electrons**; the reductant is itself **oxidised**.

Common reducing agents: reactive metals (Na , Mg , Al , Zn , Fe), hydrogen (H_2), carbon (C), Fe^{2+} , iodide (I^-), sulfite/ SO_2 and oxalate ($\text{C}_2\text{O}_4^{2-}$).

THE CLASSIC TRAP

The oxidising agent is the species that gets **reduced**; the reducing agent is the species that gets **oxidised**.

6 Standard Reduction Potentials (SRP)

What are they?

The **standard reduction potential (E°)** is a measure of the tendency of a species to be **reduced** (to gain electrons), expressed as a voltage measured against a reference half-cell.

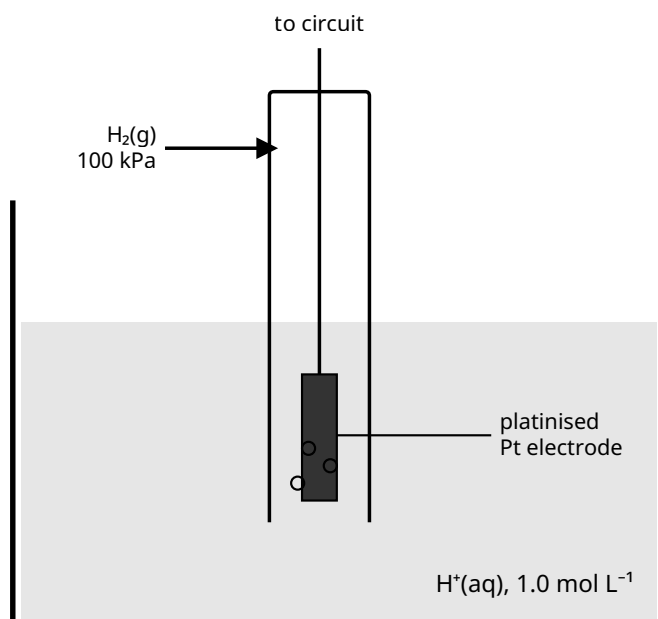
Conditions they are measured at

STANDARD CONDITIONS (THE "°")

Solution concentrations **1.0 mol L^{-1}** · gas pressure **100 kPa** · temperature **25°C (298 K)**.

What they are measured against (the reference cell)

All potentials are measured relative to the **Standard Hydrogen Electrode (SHE)**, which is **defined as exactly 0.00 V** .



$$E^\circ \equiv 0.00 \text{ V (by definition)}$$

Figure 1. The Standard Hydrogen Electrode (SHE): H_2 at 100 kPa over inert Pt in $1.0 \text{ mol L}^{-1} \text{H}^+$ at 25°C , assigned $E^\circ = 0.00 \text{ V}$ so every other half-cell can be measured against it.

How they are read from the table

READING THE SERIES

- All entries are written as **reductions**, ordered by E° .
- **More positive E°** (top) $\rightarrow$ stronger **oxidising agent** (more readily reduced), e.g. F_2 , MnO_4^- .
- **More negative E°** (bottom) $\rightarrow$ its reduced form is a stronger **reducing agent**, e.g. Li, K, Na.

How they predict whether a reaction is spontaneous

THE TEST

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode(reduction)}} - E^\circ_{\text{anode(oxidation)}}$$

A **positive** $E^\circ_{\text{cell}} \Rightarrow$ the reaction is **spontaneous**. Equivalently (the "anticlockwise rule"): the oxidant higher in the table will oxidise a reductant lower in the table.

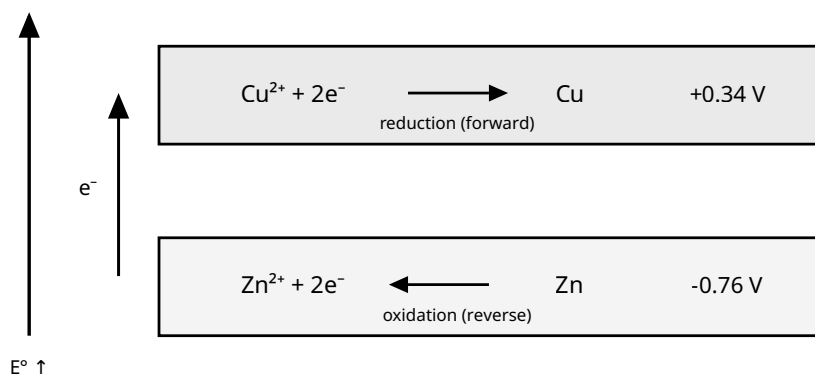


Figure 2. Predicting a spontaneous reaction. The higher- E° couple (Cu^{2+}/Cu) is reduced; the lower- E° couple (Zn^{2+}/Zn) is reversed to oxidation. $E^\circ_{\text{cell}} = (+0.34) - (-0.76) = +1.10 \text{ V} > 0$, so $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ is spontaneous.

7 Galvanic (Electrochemical) Cells

A **galvanic cell** uses a **spontaneous** redox reaction to convert chemical energy into electrical energy. The two half-reactions are separated so the electrons must travel through an external wire — this is the current.

Cell notation

Anode written on the left. | = phase boundary; || = salt bridge.

anode | anode solution || cathode solution | cathode
 e.g. **Zn(s) | Zn²⁺(aq) || Cu²⁺(aq) | Cu(s)**

Chemical principles of a galvanic cell

THE RULES (COMPLETED)

- Oxidation and reduction occur **simultaneously** but in **separate** locations.
- Oxidation occurs at the **anode** (anode & oxidation both begin with **vowels**).
- Reduction occurs at the **cathode** (cathode & reduction both begin with consonants).
- Electrons flow in the external circuit from **anode** to **cathode** (alphabetical order).
- The salt bridge maintains **electrical neutrality** (and completes the circuit; it does not carry electrons).
- Anions flow towards the **anode**.
- Cations flow towards the **cathode**.

POLARITY

In a galvanic cell the **anode is negative (–)** and the **cathode is positive (+)**.

Calculating the theoretical cell voltage

FORMULA

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

Both values taken as reductions from the SRP table. e.g. Zn|Cu cell: (+0.34) – (–0.76) = **+1.10 V**. E° is **intensive** — do not multiply it when balancing electrons.

Why the actual voltage differs from the theoretical value

CAUSES

- **Non-standard conditions** — concentrations $\neq$ 1.0 mol L^{–1}, temperature $\neq$ 25 °C, or gas pressure $\neq$ 100 kPa.
- **Internal resistance** of the cell and the **resistance of the salt bridge** cause a voltage drop when current flows.
- As the cell operates, **reactant concentrations fall** and product concentrations rise, so the voltage decreases over time.

Identifying reactions & predicting observations

HOW TO IDENTIFY EACH ELECTRODE REACTION AND THE OBSERVATIONS

- **Anodic (oxidation) reaction:** the half-cell with the **more negative** E° (lower on the series); write it **reversed** as an oxidation.
- **Cathodic (reduction) reaction:** the half-cell with the **more positive** E° (higher on the series).
- **Observations** are predicted from the half-equations: the anode electrode **dissolves / loses mass**; the cathode **gains mass** (metal deposits) or a **gas bubbles off**; solution **colours change** (e.g. blue Cu^{2+} fades, purple MnO_4^- decolourises).

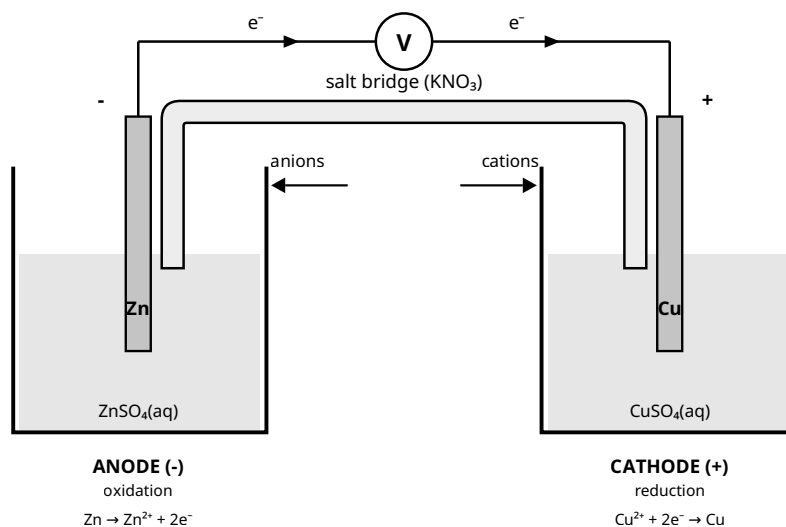


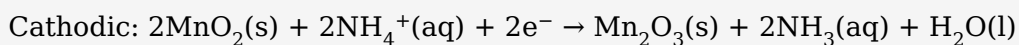
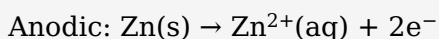
Figure 3. A standard galvanic cell (Daniell cell), annotated. Anode (-): $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ (oxidation). Cathode (+): $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (reduction). Electrons flow anode $\rightarrow$ cathode through the voltmeter; the salt bridge keeps each half-cell neutral (anions $\rightarrow$ anode, cations $\rightarrow$ cathode). Overall: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$.

8 Commercial Galvanic Cells

THREE TYPES

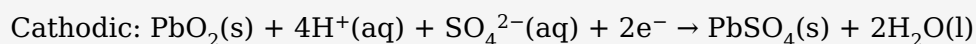
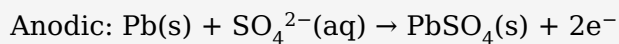
- **Primary cell** — non-rechargeable; the reaction is not reversible, so it is discarded when the reactants are used up (e.g. dry cell).
- **Secondary cell** — rechargeable; the reaction can be reversed by applying an external voltage (e.g. lead accumulator, lithium-ion).
- **Fuel cell** — the reactants (fuel + oxidant) are supplied continuously from outside the cell, so it produces electricity for as long as fuel is fed in.

Dry cell (zinc-carbon, Leclanché) — primary



Advantages	Disadvantages
Cheap, portable, made in many sizes; sealed with no free-flowing liquid.	Not rechargeable; voltage falls steadily during use; short shelf life (the zinc casing corrodes and the cell can leak).

Lead accumulator (lead-acid) — secondary



Advantages	Disadvantages
Rechargeable; delivers a large current/surge (good for starting car engines); reliable and relatively cheap; works over a wide temperature range.	Very heavy (low energy-to-mass ratio); contains toxic lead and corrosive sulfuric acid (handling/environmental hazard); damaged by repeated deep discharge.

Hydrogen-oxygen fuel cell

ACIDIC ELECTROLYTE

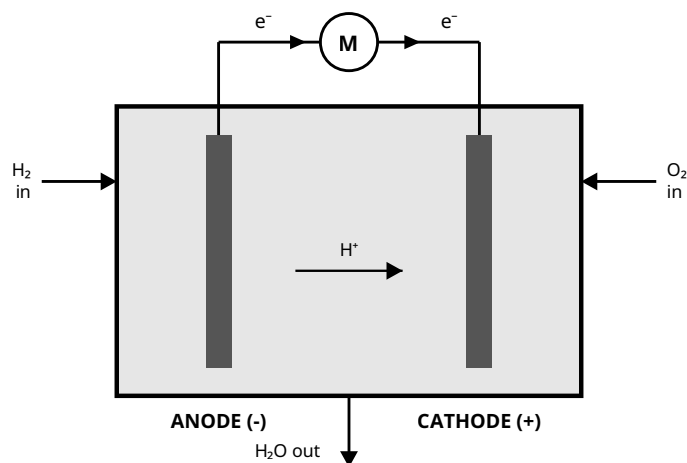
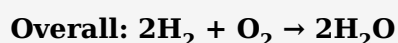


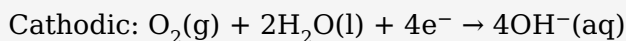
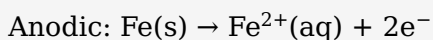
Figure 4. Hydrogen-oxygen fuel cell, annotated. Anode (–): $2\text{H}_2 \rightarrow 4\text{H}^+ + 4\text{e}^-$. Cathode (+): $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$. Electrons flow anode → cathode through the load (M); H^+ migrates through the electrolyte; the only product is water.

Advantages	Disadvantages
High efficiency (chemical energy → electricity directly, not via heat); runs continuously while fuel is supplied; only product is water (clean); quiet, no moving parts.	Expensive (platinum catalysts and materials); H_2 is hard and hazardous to store/transport (flammable, low density); H_2 is often produced from fossil fuels, so it is not always "clean".

9 Corrosion & Its Prevention

Corrosion is the unwanted oxidation of a metal by its environment. **Wet corrosion** (rusting of iron) is electrochemical and requires **both water and oxygen**; it is faster with dissolved salt or acid.

THE TWO HALF-REACTIONS OF RUSTING



Fe^{2+} is then further oxidised by O_2 to hydrated iron(III) oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ = rust.

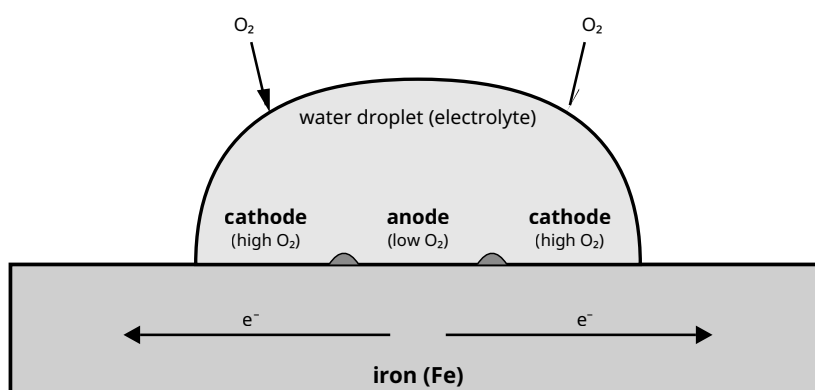


Figure 5. Wet corrosion under a water droplet. The centre (low O_2) becomes the anode where Fe is oxidised; the oxygen-rich edges become cathodes where O_2 is reduced. Fe^{2+} and OH^{-} meet and are oxidised further to deposit rust.

Methods of corrosion prevention

Method	How it works
Painting	A physical barrier that seals the metal surface so water and oxygen cannot reach it. Fails once the paint is scratched or chips off.
Oil / grease	A barrier layer (used on moving parts) that repels water and excludes oxygen; must be reapplied as it wears or washes away.
Galvanising	Coating the iron/steel with zinc . This is a barrier <i>and</i> sacrificial protection — zinc has a more negative E° (more easily oxidised), so even if the coat is scratched the zinc corrodes preferentially and protects the iron.
Cathodic protection	Forcing the iron to be the cathode so it cannot oxidise. Done either with a sacrificial anode — a more reactive metal (Mg or Zn) connected to the iron is oxidised instead — or with an impressed current from an external DC supply that feeds electrons to the iron.

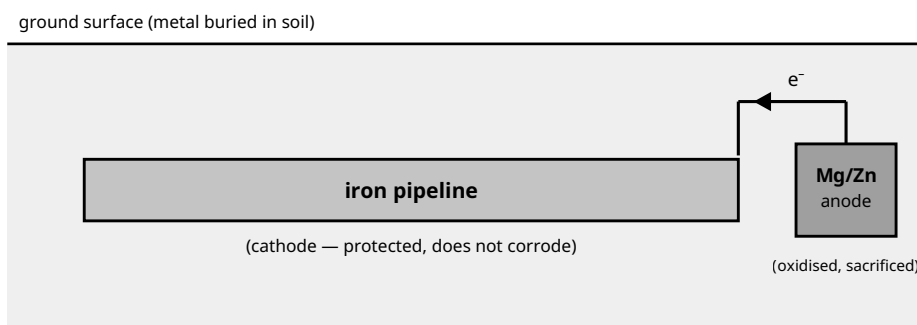


Figure 6. Cathodic (sacrificial) protection. A more reactive metal (Mg or Zn) is connected to the buried iron; the block is the anode and is oxidised ($\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$), feeding electrons into the iron so the iron becomes the cathode and cannot corrode. The sacrificial block is consumed and replaced periodically.

ELECTROLYSIS

10 Electrolysis

WHAT IS A NON-SPONTANEOUS REDOX REACTION?

A redox reaction that will **not proceed on its own** because it is energetically unfavourable (E°_{cell} is **negative**). It requires a continuous input of **electrical energy** from an external DC power supply to be driven forward. Electrolysis therefore converts **electrical** → **chemical** energy — the opposite of a galvanic cell.

WHAT STAYS THE SAME AS A GALVANIC CELL — AND WHAT FLIPS

Oxidation is **always** at the anode and reduction **always** at the cathode; electrons **always** travel anode → cathode. The **polarity flips**: in electrolysis the **anode is positive (+)** and the **cathode is negative (–)**.

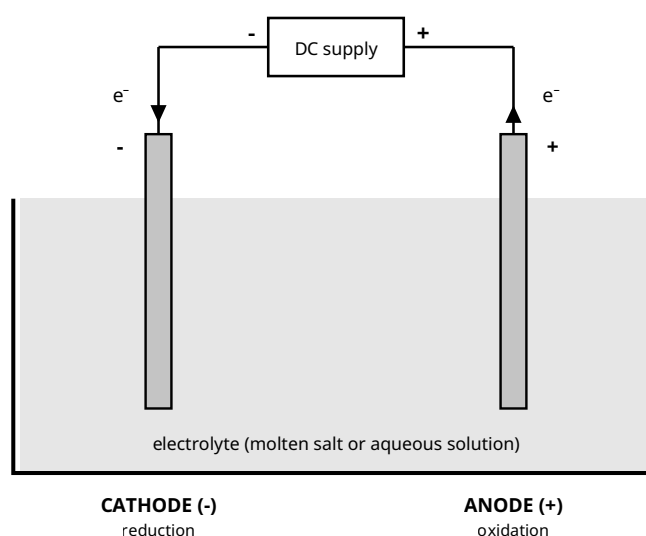


Figure 7. General electrolysis cell, annotated. The DC supply pushes electrons onto the cathode (–, reduction) and pulls them from the anode (+, oxidation). Cations move to the cathode; anions move to the anode.

Electrolysis of a molten salt (inert electrodes)

Only the ions of the salt are present, so they are discharged directly. For molten sodium chloride:

Anodic (+): $2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

Cathodic (-): $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}(\text{l})$

Electrolysis of an aqueous solution (inert electrodes)

In water, **H₂O itself** can react, competing with the dissolved ions, so there are two possibilities at each electrode:

TWO POSSIBLE ANODIC (OXIDATION) REACTIONS

the anion is oxidised — e.g. $2\text{Cl}^- \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

or water is oxidised — $2\text{H}_2\text{O} \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$ ($E^\circ = +1.23 \text{ V}$)

TWO POSSIBLE CATHODIC (REDUCTION) REACTIONS

the cation is reduced — e.g. $\text{M}^{n+} + n \text{e}^- \rightarrow \text{M}$

or water is reduced — $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-$ ($E^\circ = -0.83 \text{ V}$)

HOW THE MOST LIKELY REACTION AT EACH ELECTRODE IS CHOSEN

- **Cathode:** the species most easily reduced — the one with the **highest (most positive) E°** — reacts.
- **Anode:** the species most easily oxidised — the one with the **lowest (most negative) E°** reduction potential — reacts.
- Then allow for practical factors: **concentration** (a concentrated ion may discharge in preference to water, e.g. concentrated $\text{Cl}^- \rightarrow \text{Cl}_2$) and **overpotential** (extra voltage needed to release gases, especially O_2).

11 Applications of Electrolysis

Electroplating (e.g. silver plating)

- **Anode:** the pure plating metal — $\text{Ag}(\text{s}) \rightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$.
- **Cathode:** the object to be plated — $\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$.
- **Electrolyte:** a solution containing ions of the plating metal (e.g. silver nitrate, AgNO_3).
- **Applications:** decorative jewellery and cutlery, corrosion protection, improving appearance or surface conductivity.

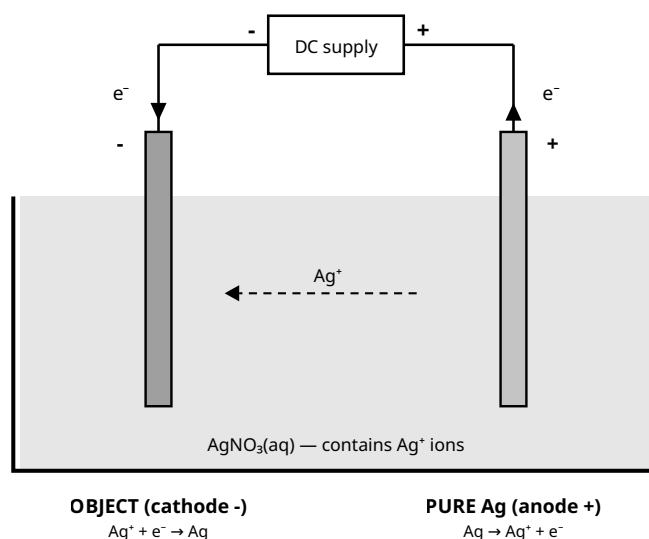


Figure 8. Silver electroplating. Ag^+ is reduced onto the object (cathode); the pure silver anode dissolves to replenish Ag^+ , keeping the solution concentration roughly constant.

Electrowinning (e.g. extraction of aluminium from alumina, Al_2O_3)

- **Anode:** $2\text{O}^{2-} \rightarrow \text{O}_2(\text{g}) + 4\text{e}^-$ (the carbon anodes then burn away: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$).
- **Cathode:** $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}(\text{l})$.
- **Electrolyte:** Al_2O_3 dissolved in molten **cryolite** (Na_3AlF_6) at $\sim 950^\circ\text{C}$, which lowers the melting point (Hall-Héroult process).

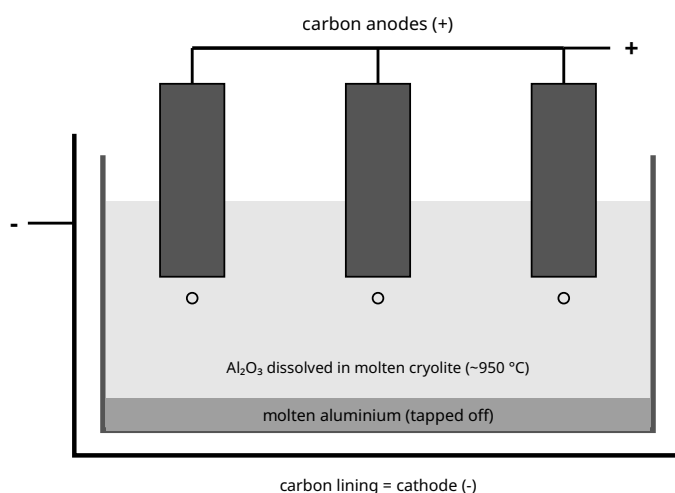


Figure 9. Extraction of aluminium (Hall-Héroult). Cathode (carbon lining): $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}(\text{l})$, collecting molten at the base. Anode (carbon blocks): $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$; the hot O_2 burns the carbon anodes, which wear away and must be replaced.

Electrorefining (e.g. purification of impure "blister" copper)

- **Anode:** the impure copper — $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ (it dissolves).
- **Cathode:** a thin sheet of pure copper — $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ (pure copper deposits).
- **Electrolyte:** acidified copper(II) sulfate solution ($\text{CuSO}_4 / \text{H}_2\text{SO}_4$).
- **Anode slime:** less-reactive impurities (Ag, Au, Pt) that do not dissolve fall below the anode as valuable "anode sludge"; more reactive impurities (e.g. Fe, Zn) stay dissolved in solution.

Recharging secondary cells (e.g. the lead accumulator)

An external supply drives the discharge reactions **in reverse** (this is electrolysis); the electrode that was the cathode on discharge becomes the anode during charging.

- **Anode (oxidation) on charge:** $\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{e}^-$.
- **Cathode (reduction) on charge:** $\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$.

Quantitative electrolysis (Faraday's laws)

METHOD

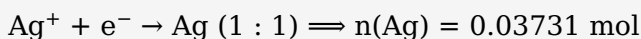
(1) $Q = I \times t$ (t in **seconds**) $\rightarrow$ (2) $n(\text{e}^-) = Q \div F$, $F = 96\,500 \text{ C mol}^{-1} \rightarrow$ (3) use the half-equation ratio to get mol of product $\rightarrow$ (4) convert to mass ($m = nM$) or gas volume.

WORKED EXAMPLE — SILVER PLATING

A current of 2.00 A flows for 30.0 min through AgNO_3 . Mass of Ag deposited?

$$Q = It = 2.00 \times (30.0 \times 60) = 3600 \text{ C}$$

$$n(\text{e}^-) = 3600 \div 96\,500 = 0.03731 \text{ mol}$$



$$m = 0.03731 \times 107.9 = \mathbf{4.03 \text{ g Ag}}$$

REFERENCE

R Standard Reduction Potentials (25 °C)

All half-equations written as reductions. Strongest oxidant at the top (most positive E°); strongest reductant at the bottom (most negative E°). Always use the exact values on your official exam data sheet.

Ordered strongest oxidant (top) $\rightarrow$ strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87	strongest oxidant
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.77	
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	+1.51	acidified permanganate
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36	
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	+1.33	acidified dichromate
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	
$\text{Br}_2(\text{l}) + 2\text{e}^- \rightarrow 2\text{Br}^-$	+1.09	
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80	
$\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$	+0.77	
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$	+0.40	
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34	
$\text{Sn}^{4+} + 2\text{e}^- \rightarrow \text{Sn}^{2+}$	+0.15	
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0.00	SHE reference
$\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13	
$\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14	

$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni(s)}$	-0.25	
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe(s)}$	-0.44	
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn(s)}$	-0.76	used in galvanising
$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-$	-0.83	
$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al(s)}$	-1.66	
$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg(s)}$	-2.37	
$\text{Na}^+ + \text{e}^- \rightarrow \text{Na(s)}$	-2.71	
$\text{Ca}^{2+} + 2\text{e}^- \rightarrow \text{Ca(s)}$	-2.87	
$\text{K}^+ + \text{e}^- \rightarrow \text{K(s)}$	-2.93	
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li(s)}$	-3.04	strongest reductant

KEY RELATIONSHIPS

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ (positive $\Rightarrow$ spontaneous). $Q = It$, $n(\text{e}^-) = Q \div F$, $F = 96\,500 \text{ C mol}^{-1}$.

Year 12 Chemistry — Redox revision summary. Half-equation balancing shown in acidic conditions only. Content written as original study notes.